Review – Integrals to Infinity

1. What does it mean for $\int_a^\infty f(x)\,dx$ to converge?

2. Does $\int_0^\infty xe^{-x^2}\,dx$ converge?

3. What does it mean for $\int_{-\infty}^b f(x)\,dx$ to converge?

4. Does $\int_{-\infty}^0 xe^{-x^2}\,dx$ converge?

5. What does it mean for $\int_{-\infty}^\infty f(x)\,dx$ to converge?

6. Does $\int_{-\infty}^\infty xe^{-x^2}\,dx$ converge?

Motivating Example

1. Evaluate $\int_{-1}^{1} \frac{1}{x^2}\, dx$ using our traditional techniques.

2. Note that $\frac{1}{x^2} > 0$ for all x. Draw the graph of $f(x) = \frac{1}{x^2}$. What can we conclude about $\int_{-1}^{1} \frac{1}{x^2}\, dx$?

3. What went wrong?

Other Improper Integrals

1. Consider $\int_{0}^{1} \frac{1}{\sqrt{x}}\, dx$. Why doesn't the Fundamental Theorem immediately apply to this integral?

2. As done previously, we can use limits to rewrite this integral:
$\int_{0}^{1} \frac{1}{\sqrt{x}}\, dx =$

3. Again, consider $\int_{-1}^{1} \frac{1}{x^2}\, dx$? How can we rewrite this integral?
$\int_{-1}^{1} \frac{1}{x^2}\, dx =$

Reciprocal Power Functions

1. Determine whether the following integrals converge or diverge:

   (a) $\displaystyle\int_1^\infty \frac{1}{x}\,dx$

   (b) $\displaystyle\int_1^\infty \frac{1}{x^{1.01}}\,dx$

   (c) $\displaystyle\int_1^\infty \frac{1}{x^{0.99}}\,dx$

2. Complete the following: $\displaystyle\int_1^\infty \frac{1}{x^p}\,dx$ converges if _____.

3. Determine whether the following integrals converge or diverge:

   (a) $\displaystyle\int_0^1 \frac{1}{x}\,dx$

   (b) $\displaystyle\int_0^1 \frac{1}{x^{1.01}}\,dx$

   (c) $\displaystyle\int_0^1 \frac{1}{x^{0.99}}\,dx$

4. Complete the following: $\int_0^1 \frac{1}{x^p}\,dx$ converges if _____.

5. Using your answers above, and noting that $\int_0^\infty f(x)\,dx = \int_0^1 f(x)\,dx + \int_1^\infty f(x)\,dx$, determine for what value(s) of p (if at all) $\int_0^\infty \frac{1}{x^p}\,dx$ converges.

Exercises

Calculate the following integrals:

1. $\int_1^e \frac{1}{x \ln x}\,dx$
   (Hint: there's a substitution in there!)

2. $\int_0^4 \frac{8}{x^2 - 16}\,dx$

   (Hint: you will likely need to show at some point that $\frac{1}{x-4} - \frac{1}{x+4} = \frac{8}{x^2-16}$. Do so.)